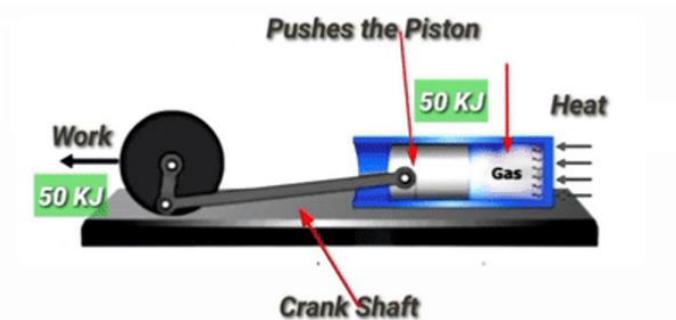


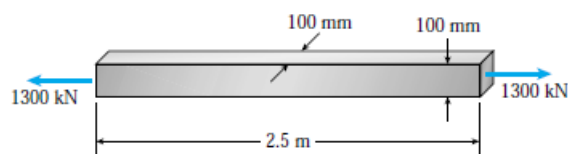
ME 150
Basics of Mechanical and Civil Engineering
Assignment#2

Instructor: Chaitanya V
Assigned Date: February 27, 2025

1. (a) In the absence of any friction and other irreversibility, can a heat engine have an efficiency of 100 percent? Explain. (2)
- (b) An automobile engine consumes fuel at a rate of 28 L/h and delivers 60 kW of power to the wheels. If the fuel has a heating value of 44,000 kJ/kg and a density of 0.8 g/cm³, determine the efficiency of this engine. (3)
- (c) Is the below engine practically possible? Explain. (2)



- (d) If the length of a wire is made double and the radius is halved of its respective values. Then, the young's modulus of the wire with respect to the initial value. (2)
- (e) A steel bar of length 2.5 m with a square cross section 100 mm on each side is subjected to an axial tensile force of 1300 kN (see figure). Assume that $E = 200$ GPa and $\nu = 0.3$. Determine the increase in volume of the bar. (3)



- (f) Source A can supply energy at a rate of 11000 kJ/min at 320°C. A second source B can supply energy at a rate of 110000 kJ/min at 68°C. Which source A or B, would you choose to supply energy to an ideal reversible engine that is to produce large amount of power if the temperature of the surroundings is 40°C? (3)
2. A reversible heat engine operates between two reservoirs at temperatures of 600°C and 40°C. The engine drives a reversible refrigerator which operates between reservoirs at temperatures of 40°C and – 20°C. The heat transfer to the heat engine is 2000 kJ and the net-work output for the combined engine refrigerator is 360 kJ. (i) Calculate the heat transfer to the refrigerant and the net heat transfer to the reservoir

at 40°C. (ii) Reconsider (i) given that the efficiency of the heat engine and the C.O.P. of the refrigerator are each 40 per cent of their maximum possible values. (10)

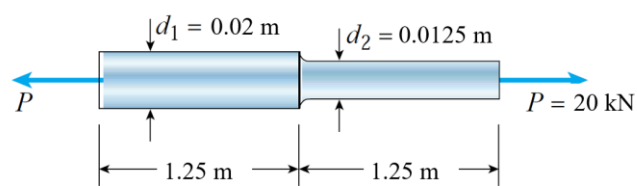
3. A reversible power cycle is used to drive a reversible heat pump cycle. The power cycle takes in Q_1 heat units at T_1 and rejects Q_2 at T_2 . The heat pump abstracts Q_4 from the sink at T_4 and discharges Q_3 at T_3 . Develop an expression for the ratio Q_4/Q_1 in terms of the four temperatures. (10)
4. A household refrigerator is maintained at a temperature of 2°C. Every time the door is opened, warm material is placed inside, introducing an average of 420 kJ, but making only a small change in the temperature of the refrigerator. The door is opened 20 times a day, and the refrigerator operates at 15% of the ideal COP. The cost of work is Rs. 2.50 per kWh. What is the monthly bill for this refrigerator? The atmosphere is at 30°C. (10)
5. A reversible engine operates between temperatures T_1 and T ($T_1 > T$). The energy rejected from this engine is received by a second reversible engine at the same temperature T . The second engine rejects energy at temperature T_2 ($T_2 < T$). Show that: (a) Temperature T is the arithmetic mean of temperatures T_1 and T_2 if the engines produce the same amount of work output (b) Temperature T is the geometric mean of temperatures T_1 and T_2 if the engines have the same cycle efficiencies. (15)
6. On the basis of the first law fill in the blank spaces in the following table of imaginary heat engine cycles. On the basis of the second law classify each cycle as reversible, irreversible, or impossible. (10)

	Cycle Temperature		Rate of Heat flow		Rate of work output	Efficiency
	Source	Sink	Supply	Rejection		
(a)	327°C	27°C	420kJ/s	230 kJ/s	()kW	()
(b)	1000°C	100°C	()kJ/s	4.2 MJ/min	()kW	65%
(c)	750K	300K	()kJ/s	()kJ/s	26kW	60%
(d)	700K	300K	3kW	()kW	2kW	()

7. A circular bar is rigidly fixed at both ends, where both ends are uniformly taper from 50 mm at one end to 30 mm at the other end. If temperature is raised through 30K, find the maximum stress developed in the bar. Take E as 120GPa and α as 6×10^{-6} /K for the material. (10)
8. The true stress (in MPa) versus true strain relationship for a metal is given by:

$\sigma = A\varepsilon^n$, where $A = 1020$ and $n=0.4$. The cross-sectional area at the start of a test (when the stress and strain values are equal to zero) is 100 mm^2 . Calculate the cross-sectional area at the time of necking (in mm^2). At the initiation of the necking $\varepsilon = n$. Find the engineering stress and strain at the time of necking. (10)

9. Steel bar 2.5 m long has a circular cross section of diameter $d_1 = 0.02 \text{ m}$ over one-half of its length and diameter $d_2 = 0.0125 \text{ m}$ over the other half. The modulus of elasticity $E = 200 \text{ GPa}$. (a) How much will the bar elongate under a tensile load $P = 20 \text{ kN}$? (b) If the same volume of material is made into a bar of constant diameter d and length 2.5 m, what will be the elongation under the same load P ? (10)



10. (Bonus) Below Fig. shows a rigid bar hinged at A and supported in a horizontal position by two vertical identical steel wires. Neglect the weight of the beam. The tension T_1 and T_2 induced in these wires by a vertical load P applied as shown are (10)

